

## Spring Boot Backend

*Transit KMS Client Service*

- Endpoint: `http://haproxy:8200`
- AES Key Wrapping & Unwrapping
- KMS Shredding & RSA-2048 Proof Signing
- Tombstone Reconciler & Memory Zeroize

## HAProxy Load Balancer

*Port :8200 (Single Gateway Entrypoint)*

- Health Poller: `GET /v1/sys/health` (500ms)
- HTTP 200 OK → Route to Active Leader
- HTTP 429 → Bypass Standby Followers
- Zero-Downtime Auto-Failover (< 2.5s)

## HashiCorp Vault 3-Node Raft Cluster

*Integrated Storage WAL | Quorum: 2/3 Nodes*

Active Traffic (HTTP 200)

Health Poll: 429

### Vault Node 2 (vault-2)

STANDBY

*Hot Standby Follower (:8200)*

- Raft Quorum Voter | Append-Only WAL Sync
- Health: HTTP 429 (Standby Replica)

### Vault Node 3 (vault-3)

STANDBY

*Hot Standby Follower (:8200)*

- Raft Quorum Voter | Append-Only WAL Sync
- Health: HTTP 429 (Standby Replica)

### Vault Node 1 (vault-1)

ACTIVE LEADER

*Active Raft Leader (:8200)*

- Transit Secrets Engine Mounted
- Master KEK Lifecycle & Destruction
- RSA-2048 Proof Signing Authority
- Health: HTTP 200 OK (Active Leader)
- Direct Master Key Wrap/Unwrap

Raft Linearizable WAL Replication — Heartbeats 250ms | AppendEntries :8201

## Automated Disaster Recovery & Failover Workflow (< 2.5s)

### 1. Heartbeat Loss

vault-1 crashes.  
Followers detect  
timeout (>500ms).

### 2. Raft Election

vault-2 starts T+1.  
Secures 2/3 quorum,  
re-elects in 1.84s.

### 3. State Promotion

vault-2 mounts  
transit engine,  
returns HTTP 200.

### 4. Traffic Reroute

HAProxy detects  
health in 2.15s,  
failover < 2.5s.